

AISHWARIYA A

AI / ML Engineer

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PROFESSIONAL SUMMARY

Motivated AI/ML Engineer (Fresher) with hands-on experience building end-to-end machine learning, deep learning, and Generative AI pipelines. Proficient in NLP, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), computer vision, and multimodal systems. Skilled in model development, training, evaluation, and deployment using industry-standard frameworks. Eager to contribute to AI/ML product teams and solve real-world problems through data-driven solutions.

TECHNICAL SKILLS

Programming Languages	Python, SQL
Machine Learning	Supervised & Unsupervised Learning, Classification, Regression, Clustering, Feature Engineering, Hyperparameter Tuning, Model Evaluation
Deep Learning	ANN, CNN, RNN, LSTM, Autoencoders, Transfer Learning
NLP & Generative AI	Tokenization, Word Embeddings, Transformers, LLMs, Prompt Engineering, RAG, Vector Databases, Semantic Search
Frameworks & Tools	PyTorch, TensorFlow, Scikit-learn, Hugging Face, LangChain, LangGraph, FAISS
Deployment & DevOps	Docker, FastAPI, Git, GitHub
Data Visualization	Power BI, Tableau, Matplotlib, Seaborn

INTERNSHIP EXPERIENCE

Robotics Intern — Flo Mobility, Bangalore

Jun 2024 – Jul 2024

- Performed data annotation and labelling for construction site imagery using Roboflow, preparing training datasets for object detection models used in building construction robots.
- Gained hands-on experience with PCB design fundamentals, supporting hardware-software integration in autonomous robotic systems.
- Collaborated on CAD modelling tasks supporting the mechanical design of construction robots.

PROJECTS

Healthcare Retrieval-Augmented Generation (RAG) System

- Engineered a RAG pipeline ingesting and chunking medical PDFs, generating dense embeddings indexed via FAISS for efficient semantic retrieval.
- Improved factual accuracy and retrieval relevance over keyword-based baselines through embedding-driven similarity matching and re-ranking.
- Implemented safety mechanisms — source filtering, deduplication, context-limited prompting — reducing LLM hallucinations in medical outputs.

Tech Stack: Python · FAISS · Hugging Face Transformers · LangChain

Multimodal Anomaly Detection System

- Designed a framework fusing sensor time-series, industrial images, and system logs for fault detection in manufacturing environments.
- Built LSTM autoencoder (temporal), CNN autoencoder (visual), and log-based scorer unified into a single decision framework; reduced false positives by 30%+.
- Evaluated using ROC-AUC, Precision, Recall, and F1-Score across all modalities.

Tech Stack: Python · PyTorch · TensorFlow · LSTM · CNN · Scikit-learn

End-to-End Customer Churn Prediction

- Built a full ML pipeline — EDA, data cleaning, feature engineering, SMOTE balancing, model training, and evaluation using Logistic Regression and ensemble methods.
- Achieved superior churn identification over rule-based baselines via cross-validation, AUC-ROC analysis, and hyperparameter tuning.

Tech Stack: Python · Pandas · Scikit-learn · Matplotlib · Seaborn

EDUCATION

Bachelor of Engineering — Robotics and Automation

2021 – 2025

Sri Ramakrishna Engineering College, Coimbatore | CGPA: 7.94 / 10

CERTIFICATIONS

- Master Program in Data Science and Artificial Intelligence — FITA Academy, Chennai